UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Particular Integral when X is Any Other Form

Complete study pack for class 009 (the deck's SESSION 7): when X is none of the four standard forms, factorise $f(D)$ into linear factors, split $\frac{1}{f(D)}$ into partial fractions and apply $\frac{1}{D-m}X = e^{mx}\int Xe^{-mx}\,dx$ to each term; worked for $\tan ax$, $\ln x$, e^{e^x}, $\sin(e^x)$ and $e^{2x}\sin(e^x)$, with the one ESA question solved in full and every line checked with sympy

Compiled from the PESU Unit 2 class 009 slide deck (18 slides, title slide “SESSION: 7”), its two AV-summary transcripts (T009, 15.2 min; T009b “Continued”, 10.2 min), the Unit 1 & 2 class-work sheet (as a pointer only) and eight past ESA papers. Generated 16 October 2026.

· How to use this pack

Topic 1 is the method itself (Figures F1.1, F1.2): the inverse operator $\frac{1}{D-m}$ as an integral, partial fractions, and the shortcut of applying the linear factors one after another without partial fractions; the example $y'' = \ln x$ is the case $m = 0$ (integrate twice). Topic 2 is $(D^2 + a^2)y = \tan ax$, where the factors are complex ($D \pm ia$). Topic 3 is the substitution $t = e^x$ for right-hand sides built from e^x (e^{e^x}, $\sin(e^x)$, $e^{2x}\sin(e^x)$), including the ESA question (W3.1). CF cases are in the class 002 pack; the four standard forms (Types 1–5: e^{ax}, $\sin$ / $\cos$, x^m, $e^{ax}V$, xV) in the class 003–006 packs. They are not re-taught here, but every worked example is complete. Right-hand sides $\tan ax$, $\sec ax$, $\csc ax$ can also be done by variation of parameters (class 011 VoP pack; CW P23, P26, P28 ask for that method) — see C2.1. Every topic has the same parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S009·5	Unit 2 deck 009 (“Particular integral when X is any other form”, SESSION 7), slide 5 (rendered .pptx)
T009 00:03:13	Unit 2 AV-summary transcript “Particular Integral when X is any other form” (15.2 min), segment start
T009b 00:01:10	Unit 2 AV-summary transcript “Particular integral when X is any other form_Continued” (10.2 min), segment start
CW U2 P28	Unit 2 class-work problem 28 (pointer only; solved in the class 011 VoP pack)
ESA Dec-23 3c	Past paper: UE22MA141A Dec 2023, question 3.c

Contents

1. 1 · The general method: linear factors, partial fractions and $\frac{1}{D-m}X$
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3. 3 · Right-hand sides in e^x: the substitution $t = e^x$
4. Appendix A · Symbol glossary
5. Appendix B · Sources, coverage and gaps

1 The general method: linear factors, partial fractions and $\frac{1}{D-m}X$

When the right-hand side X is not e^{ax}, $\sin(ax+b)$ or $\cos(ax+b)$, a polynomial x^m, or $e^{ax}V$, none of the shortcut rules applies. Then $f(D)$ is resolved into linear factors $(D-m_1)\cdots(D-m_n)$, $\frac{1}{f(D)}$ is split into partial fractions $\sum \frac{A_k}{D-m_k}$, and each term is evaluated by the standard result $\frac{1}{D-m}X = e^{mx}\int Xe^{-mx}\,dx$. The CF is found exactly as before. With $m=0$ the result is plain integration, so $\frac{1}{D^2}\ln x$ is $\ln x$ integrated twice (slides 009·8–9).

Key terms and definitions

"Any other form". A right-hand side X that is not one of the four standard forms: not e^{ax}, not $\sin(ax+b)$ or $\cos(ax+b)$, not a polynomial x^m, and not $e^{ax}V$. Examples: $\tan ax$, $\ln x$, e^{e^x}, $\sin(e^x)$. S009·3 · T009 00:00:07 · T009 00:00:27 · T009b 00:00:32

Linear factors of $f(D)$. $f(D) = (D-m_1)(D-m_2)\cdots(D-m_n)$, where $m_1,\dots,m_n$ are the roots of the auxiliary equation $f(m)=0$ (real or complex). S009·3 · T009 00:01:19

Inverse operator $\frac{1}{D-m}$. $\frac{1}{D-m}X$ is the function u with $(D-m)u = X$, i.e. $u' - mu = X$; it equals $e^{mx}\int Xe^{-mx}\,dx$ (no constant of integration: the constants are in the CF). S009·4 · T009 00:02:11 · T009b 00:03:01

$\frac{1}{D^k}$. k successive integrations with respect to x (the case $m=0$ of $\frac{1}{D-m}$, applied k times). S009·8 · T009 00:12:09

> *"When X is any other function of x. Rules for Finding the Particular Integral. Consider the differential equation, $(D^n + a_1D^{n-1} + a_2D^{n-2} + \cdots + a_{n-1}D + a_n)y = X$. It can be written as: $f(D)y = X$ ∴ P.I. $= \frac{1}{f(D)}X$. Resolve $f(D)$ into linear factors. Let $f(D) = (D-m_1)(D-m_2)\dots(D-m_n)$. Then, P.I. $= \frac{1}{f(D)}X = \frac{1}{(D-m_1)(D-m_2)\dots(D-m_n)}X$."* S009·3

> *"Use partial fractions: $P.I = \left(\frac{A_1}{D-m_1} + \frac{A_2}{D-m_2} + \cdots + \frac{A_n}{D-m_n}\right)X$, $P.I = A_1\frac{1}{D-m_1}X + A_2\frac{1}{D-m_2}X + \cdots + A_n\frac{1}{D-m_n}X$, $P.I = A_1e^{m_1x}\int Xe^{-m_1x}dx + A_2e^{m_2x}\int Xe^{-m_2x}dx + \cdots + A_ne^{m_nx}\int Xe^{-m_nx}dx$."* S009·4

> *"Standard formula (lecture): "one over D minus a of $f(x)$ … is nothing but e^{ax} integral $e^{-ax}f(x)\,dx$"."* T009 00:02:11 · T009b 00:03:01

Formulas

E1.1 · The standard result for one linear factor ESA ★★

$$\frac{1}{D-m}\,X = e^{mx}\int X\,e^{-mx}\,dx$$

GIVES The PI contribution of one factor $D-m$.

APPLIES Any X; m real or complex.

NOTE Sign: $\frac{1}{D+1}X = e^{-x}\int Xe^{x}dx$ ($m=-1$). No constant of integration.

S009·4 · T009 00:02:11 · T009b 00:03:01 · T009b 00:03:36 · ESA Dec-23 3c

E1.2 · Why: a first-order linear equation ESA ★

$$u = \frac{1}{D-m}X \iff u' - mu = X \iff \left(e^{-mx}u\right)' = Xe^{-mx}$$

GIVES E1.1 by the integrating factor e^{-mx}.

APPLIES Derivation; not on the slides.

NOTE Integrating and multiplying by e^{mx} gives E1.1.

ADDED FOR CLARITY S009·4

E1.3 · Partial fractions of $\frac{1}{f(D)}$ ESA ★★

$$\frac{1}{f(D)} = \frac{A_1}{D-m_1} + \cdots + \frac{A_n}{D-m_n}$$

$$\text{P.I.} = A_1e^{m_1x}\int Xe^{-m_1x}dx + \cdots + A_ne^{m_nx}\int Xe^{-m_nx}dx$$

GIVES The PI as a sum of n single-factor terms.

APPLIES Distinct roots $m_1,\ldots,m_n$.

NOTE Cover-up: $A_k = \frac{1}{f'(m_k)}$; e.g. $\frac{1}{(D+1)(D+2)} = \frac{1}{D+1} - \frac{1}{D+2}$.

S009·4 · S009·10 · S009·12 · S009·14 · T009 00:01:32 · T009 00:01:52

E1.4 · Successive application (no partial fractions) ESA ★★

$$\frac{1}{(D-m_1)(D-m_2)}X = \frac{1}{D-m_1}\left[\frac{1}{D-m_2}X\right] = \frac{1}{D-m_1}\left[e^{m_2x}\int Xe^{-m_2x}dx\right]$$

GIVES The same PI by two single-factor steps, inner factor first.

APPLIES Also with repeated roots ($(D-m)^2$), where partial fractions need an extra term.

NOTE The factors commute; choose the order that makes the integrals easy (W3.1, W3.2).

ADDED FOR CLARITY S009·3 · S009·10 · ESA Dec-23 3c

E1.5 · $m = 0$: integrate

ESA ★

$$\frac{1}{D}X = \int X\,dx, \qquad \frac{1}{D^2}X = \iint X\,dx\,dx$$

GIVES The PI when $f(D) = D^k$.

APPLIES $y'' = X$, $y''' = X$, ...

NOTE $\iint \ln x\,dx\,dx = \frac{x^2 \ln x}{2} - \frac{3x^2}{4}$ (slide 009·9).

S009·8 · S009·9 · T009 00:12:09

E1.6 · Integrals of $\ln x$

ESA ★

$$\int \ln x\,dx = x\ln x - x, \qquad \int x\ln x\,dx = \frac{x^2}{2}\ln x - \frac{x^2}{4}$$

GIVES The two integrations of slide 009·9.

APPLIES By parts with $u = \ln x$.

NOTE Write $\ln x = \ln x \cdot 1$ for the first one.

S009·9 · T009 00:12:37

Graphs and diagrams

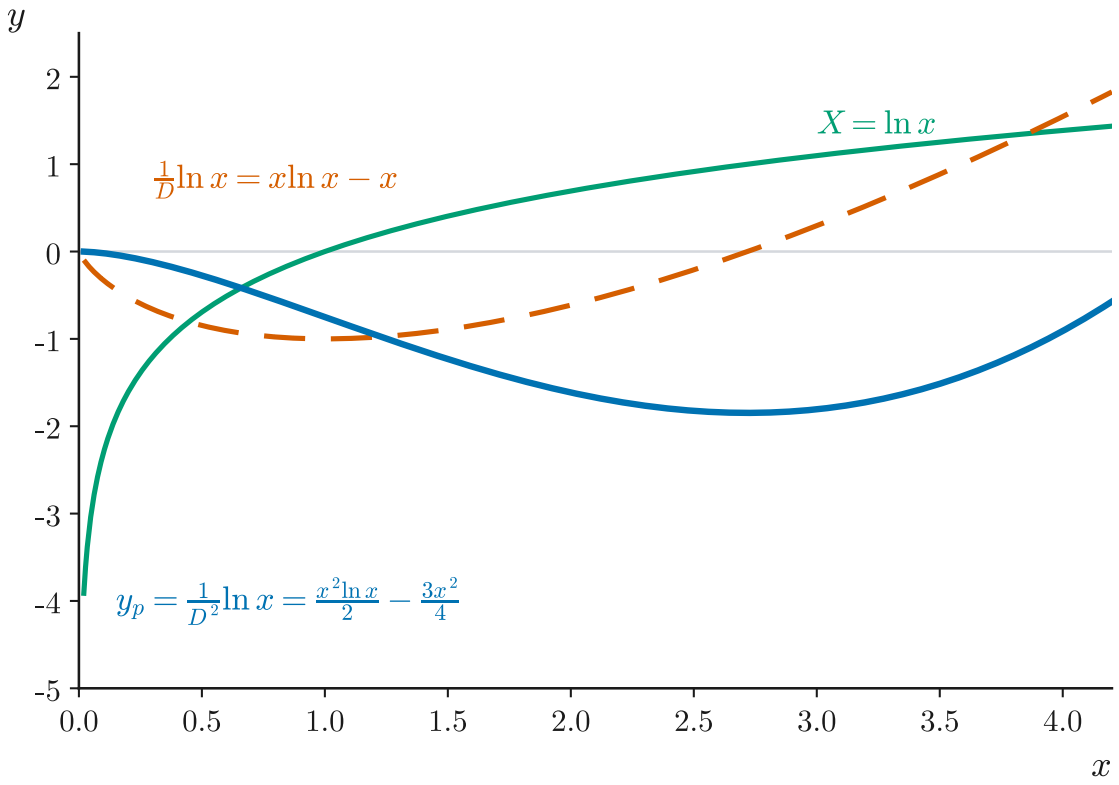
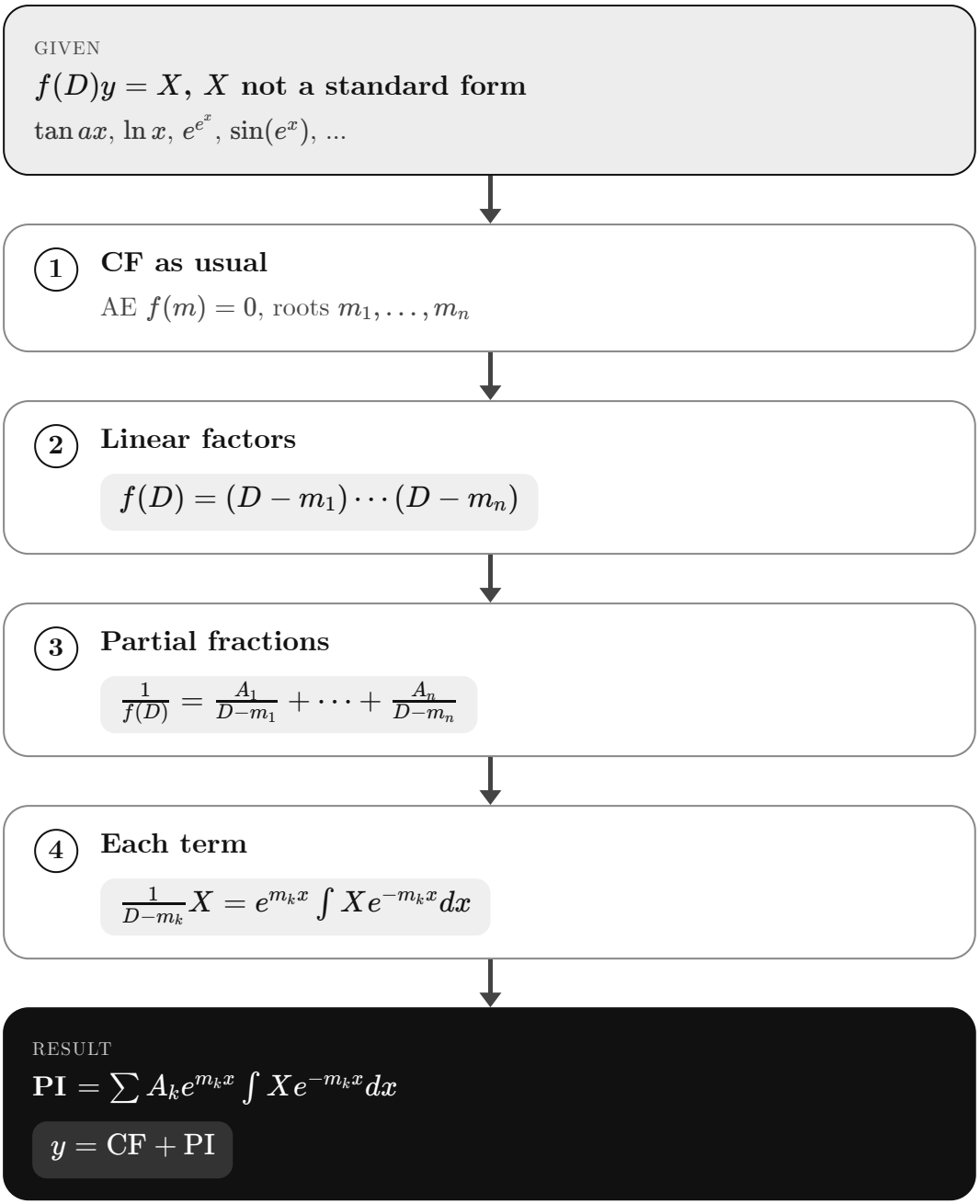


Figure G1.1. $\frac{1}{D^2}\ln x$: integrate twice. ESA ★

ADDED FOR CLARITY S009·8 · S009·9

How to read it. Green: the right-hand side $\ln x$. Vermilion dashed: one integration, $\frac{1}{D}\ln x = x\ln x - x$ (its slope is $\ln x$). Blue: the second integration, the PI $\frac{x^2\ln x}{2} - \frac{3x^2}{4}$, whose slope is the vermilion curve and whose curvature is the green one. Each curve has a turning point where the one below it crosses zero ($x = 1$ for $x\ln x - x$, $x = e$ for the PI).

Flowcharts

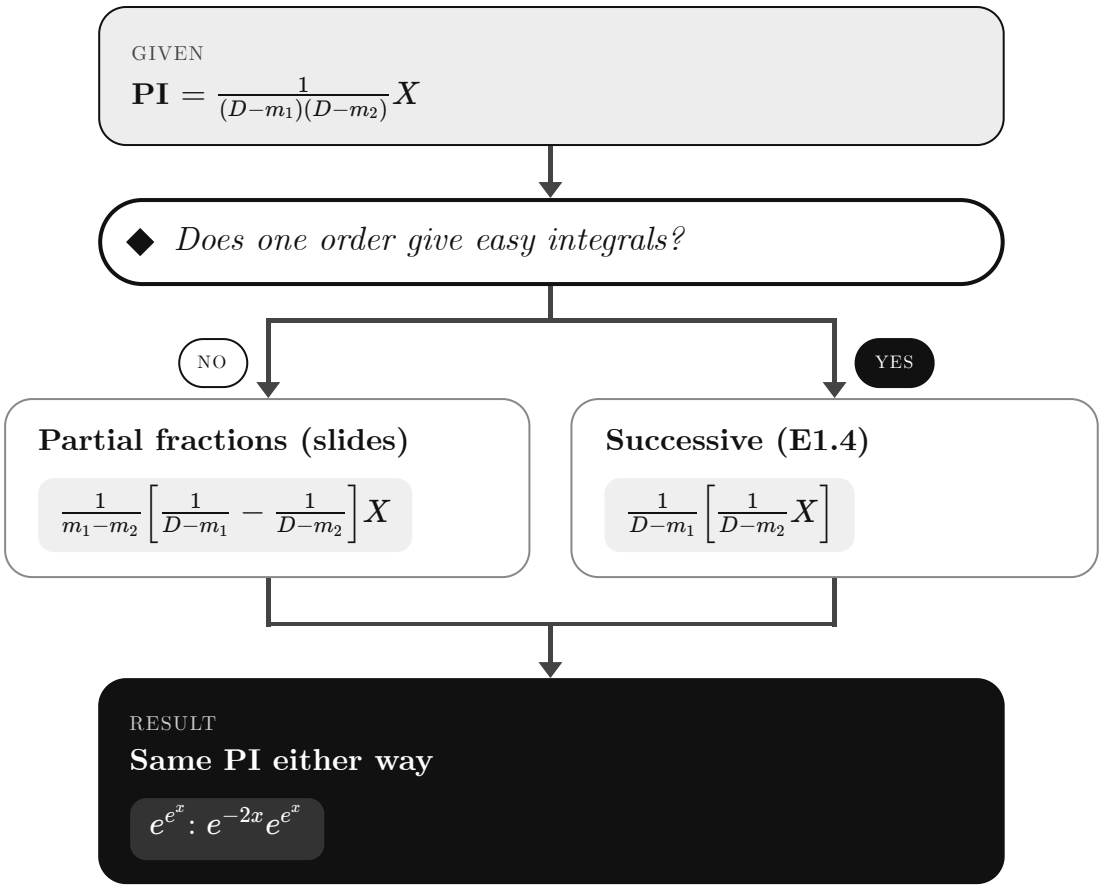


Remark. No constants of integration in step 4; the CF already holds them.

Exam trap. $\frac{1}{D+2}$ has $m = -2$: outside e^{-2x}, inside e^{+2x}.

Figure F1.1. PI when X is any other form (slides 009·3–4) ESA ★★

S009·3 · S009·4 · T009 00:00:58 · T009 00:01:32 · T009 00:02:11



Remark. The slides always use partial fractions; the successive route is a check or a shortcut. For a repeated factor $(D-m)^2$ use the successive route.

Figure F1.2. Partial fractions or one factor at a time? ESA ★★

ADDED FOR CLARITY S009·4 · S009·10 · ESA Dec-23 3c

Which PI rule? (the four standard forms and "any other form")

X	Rule	Class pack
e^{ax}	$D \to a$	003
$\sin(ax+b)$, $\cos(ax+b)$	$D^2 \to -a^2$	004
x^m	series in D up to D^m	005
$e^{ax}V$ **(and** xV**)**	$e^{ax}\frac{1}{f(D+a)}V$; $\left[\frac{x}{f(D)} - \frac{f'(D)}{(f(D))^2}\right]V$	006
anything else	linear factors, partial fractions, $\frac{1}{D-m}X = e^{mx}\int Xe^{-mx}dx$	009 (this pack)

T009 00:00:27 · T009b 00:00:32 · S009·3

Important points and why

1. First check the four standard forms (e^{ax}; $\sin/\cos(ax+b)$; x^m; $e^{ax}V$). Only if X is none of them ($\tan ax$, $\ln x$, e^{e^x}, $\sin(e^x)$) use this method. ESA ★★ T009 00:00:27 · T009 00:03:13 · T009b 00:00:32 · T009b 00:01:25
2. "There's no change in computing the complementary function": AE, roots, CF exactly as in class 002. ESA ★★ T009 00:03:33 · S009·5
3. Procedure: factorise $f(D)$ into linear factors → partial fractions $\sum \frac{A_k}{D-m_k}$ → $\frac{1}{D-m}X = e^{mx}\int Xe^{-mx}dx$ for each term → add (Figure F1.1). ESA ★★ S009·3 · S009·4 · T009 00:01:19 · T009 00:01:52 · T009 00:02:11
4. Watch the sign: $\frac{1}{D+2}X = e^{-2x}\int Xe^{2x}dx$ ($m=-2$): the outside exponential has the AE root, the inside one its negative. ESA ★★ T009b 00:03:36 · S009·11 · S009·12
5. No constants of integration in a PI: the arbitrary constants are already in the CF. ESA ★★ ADDED FOR CLARITY S009·9 · S009·11
6. With $m=0$ ($f(D)=D^2$), $\frac{1}{D^2}$ means "integrate twice": $\frac{1}{D^2}\ln x = \frac{x^2\ln x}{2} - \frac{3x^2}{4}$. ESA ★ S009·8 · S009·9 · T009 00:12:09
7. Shortcut (not on the slides): apply the factors one after another, $\frac{1}{D-m_1}\left[\frac{1}{D-m_2}X\right]$; this avoids partial fractions and also works for repeated factors. ESA ★★ ADDED FOR CLARITY S009·3 · ESA Dec-23 3c

Why does $\frac{1}{D-m}X$ equal $e^{mx}\int Xe^{-mx}dx$?
$u = \frac{1}{D-m}X$ means $u' - mu = X$, a first-order linear equation. Multiplying by the integrating factor e^{-mx} gives $(ue^{-mx})' = Xe^{-mx}$; integrating and multiplying by e^{mx} gives the formula.
ADDED FOR CLARITY S009·4

Why do the shortcut rules of Types 1–5 fail for $\tan ax$ or e^{e^x}?
Those rules work because D maps e^{ax}, $\sin ax$, x^m into the same family. The derivatives of $\tan ax$ or e^{e^x} are new functions every time, so no substitution $D \to a$ or $D^2 \to -a^2$ exists; the integral formula always works.
ADDED FOR CLARITY T009 00:00:27 · T009 00:03:13

Exam trap. Do not write $\frac{1}{D+1}X = e^x\int Xe^{-x}dx$: for $D+1 = D-(-1)$, $m=-1$, so it is $e^{-x}\int Xe^{x}dx$. ADDED FOR CLARITY T009b 00:03:36 · S009·12

Exam trap. Partial fractions with the wrong sign: $\frac{1}{(D+1)(D+2)} = \frac{1}{D+1} - \frac{1}{D+2}$ ($A=1$ at $D=-1$, $B=-1$ at $D=-2$). Slide 009·10 lists the fractions as $\frac{A}{D+2} + \frac{B}{D+1}$, so there $A=-1$, $B=1$ — the same split. S009·10 · S009·12

Exam trap. Do not forget the CF: "solve" means $y = y_c + y_p$ (slides 009·10–17 stop at the PI; the ESA question asks to solve). ADDED FOR CLARITY ESA Dec-23 3c · S009·11

Worked examples

W1.1 · $\frac{d^2y}{dx^2} = \ln x$ (slides 009·8–9) ESA ★

GIVEN Solve $\frac{d^2y}{dx^2} = \ln x$, i.e. $D^2y = \ln x$.

STEP 1 AE $m^2 = 0$: $m = 0, 0$. CF $= c_1 + c_2x$.

STEP 2 PI $= \frac{1}{D^2}\ln x = \iint \ln x\,dx\,dx$ ($\frac{1}{D-m}$ with $m = 0$, twice).

STEP 3 First: $\int \ln x \cdot 1\,dx = x\ln x - \int x \cdot \frac{1}{x}\,dx = x\ln x - x$.

STEP 4 Second: $\int (x\ln x - x)\,dx = \frac{x^2}{2}\ln x - \frac{1}{2}\int x\,dx - \int x\,dx = \frac{x^2}{2}\ln x - \frac{x^2}{4} - \frac{x^2}{2}$.

ANSWER $$\boxed{y = c_1 + c_2x + \frac{x^2\ln x}{2} - \frac{3x^2}{4}}$$

CHECK Slide answer agrees; sympy dsolve gives the same, and $\frac{d^2}{dx^2}\left(\frac{x^2\ln x}{2} - \frac{3x^2}{4}\right) = \ln x$. Figure G1.1.

S009·8 · S009·9 · T009 00:11:30 · T009 00:12:55

SHORT-NOTE RECAP

When X is not e^{ax}, $\sin/\cos(ax+b)$, x^m or $e^{ax}V$: CF as usual; factorise $f(D) = (D - m_1)\cdots(D - m_n)$; partial fractions $\frac{1}{f(D)} = \sum \frac{A_k}{D-m_k}$; each term by $\frac{1}{D-m}X = e^{mx}\int Xe^{-mx}dx$ (no constants). Or apply the factors one after another. $m = 0$: integrate. Example: $D^2y = \ln x$: $y = c_1 + c_2x + \frac{x^2\ln x}{2} - \frac{3x^2}{4}$.

2 Complex linear factors: $(D^2+a^2)y=\tan ax$

For $D^2+a^2=(D+ia)(D-ia)$ the partial fractions have complex coefficients, $\frac{1}{2ia}\left[\frac{1}{D-ia}-\frac{1}{D+ia}\right]$. Each term is an integral of $\tan ax\, e^{\mp iax}$, which Euler's formula $e^{\mp iax}=\cos ax \mp i\sin ax$ turns into elementary integrals; the second term follows from the first by changing i to $-i$. The imaginary parts cancel and the PI is real: $-\frac{1}{a^2}\cos ax\log(\sec ax+\tan ax)$. The same equation can also be solved by variation of parameters (class 011).

Formulas

E2.1 · Complex factors of D^2+a^2 ESA ★★

$$\frac{1}{D^2+a^2}=\frac{1}{(D+ia)(D-ia)}=\frac{1}{2ia}\left[\frac{1}{D-ia}-\frac{1}{D+ia}\right]$$

GIVES The partial fractions for complex AE roots $\pm ia$.

APPLIES CF $c_1\cos ax+c_2\sin ax$.

NOTE $A=\frac{1}{2ia}$, $B=-\frac{1}{2ia}$ (T009 00:05:02).

S009·5 · S009·6 · T009 00:04:31

E2.2 · Euler's formula ESA ★

$$e^{\pm iax}=\cos ax\pm i\sin ax$$

GIVES $e^{-iax}\tan ax=\sin ax-i\frac{\sin^2 ax}{\cos ax}$, and $e^{iax}(\cos ax-i\sin ax)=1$.

APPLIES Inside $\int \tan ax\, e^{\mp iax}dx$.

NOTE $\frac{\sin^2 ax}{\cos ax}=\sec ax-\cos ax$.

S009·6 · T009 00:05:33 · T009 00:07:03

E2.3 · Integrals used ESA ★

$$\int \sin ax\,dx=-\frac{\cos ax}{a},\qquad \int \sec ax\,dx=\frac{1}{a}\log(\sec ax+\tan ax)$$

GIVES The two pieces of $\int \tan ax\, e^{-iax}dx$.

APPLIES Slide 009·6.

NOTE Also $\int \cos ax\,dx=\frac{\sin ax}{a}$.

S009·6 · T009 00:07:38

E2.4 · One factor on $\tan ax$ ESA ★★

$$\frac{1}{D \mp ia}\tan ax = -\frac{1}{a}\left[1 \pm i\, e^{\pm iax}\log(\sec ax + \tan ax)\right]$$

GIVES The two partial results (upper signs: slide 009·6; lower signs: 009·7 by $i \to -i$).

APPLIES $(D^2 + a^2)y = \tan ax$.

NOTE Changing i to $-i$ gives the second from the first.

S009·6 · S009·7 · T009 00:08:41 · T009 00:08:58

E2.5 · Result: PI of $\tan ax$ ESA ★★

$$\frac{1}{D^2 + a^2}\tan ax = -\frac{1}{a^2}\cos ax\,\log(\sec ax + \tan ax)$$

GIVES The PI; with the CF, $y = c_1\cos ax + c_2\sin ax - \frac{1}{a^2}\cos ax\log(\sec ax + \tan ax)$.

APPLIES Any $a \neq 0$.

NOTE $a = 2$ gives CW P28's answer (asked there by variation of parameters).

S009·7 · T009 00:11:09 · CW U2 P28

Graphs and diagrams

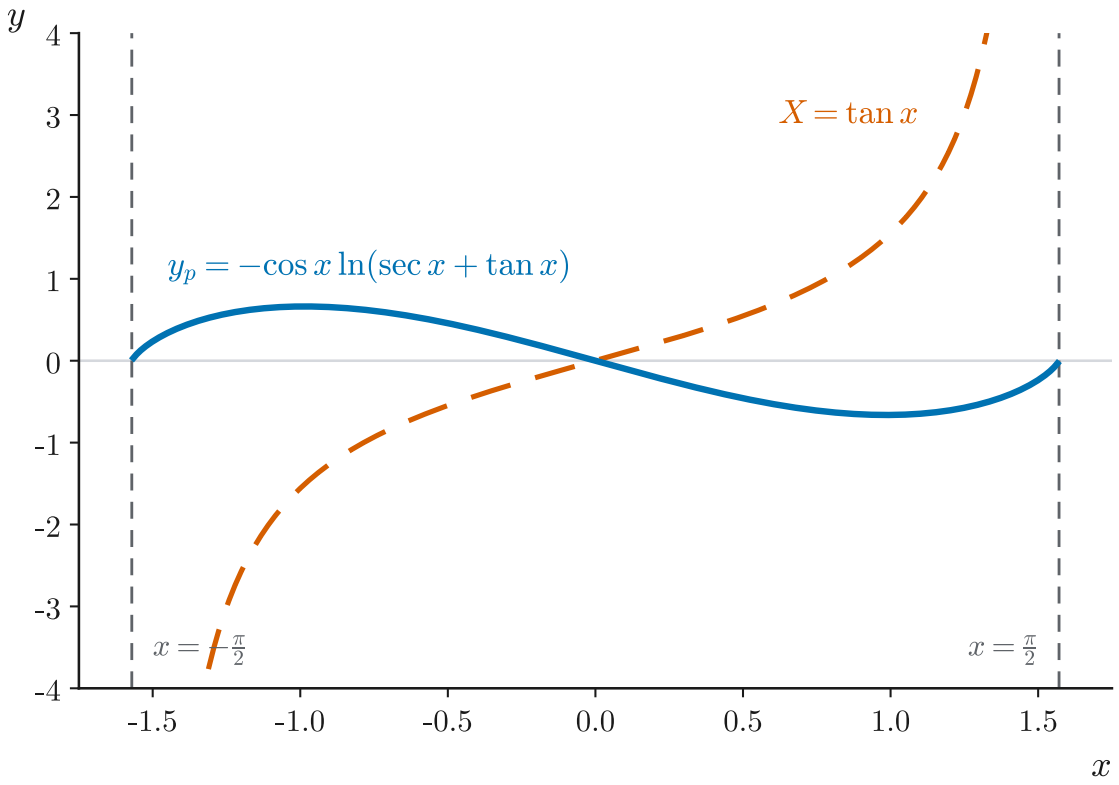
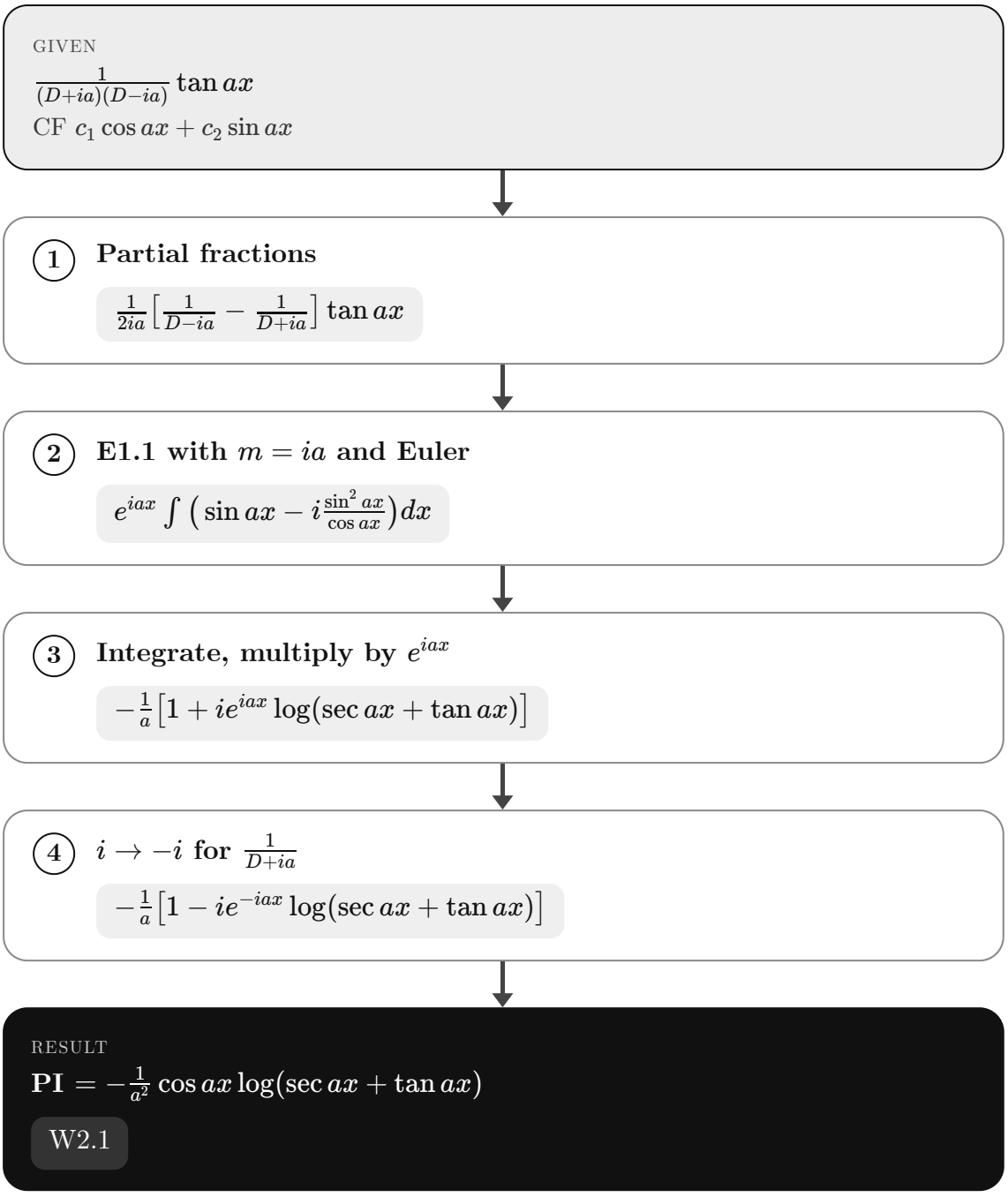


Figure G2.1. $(D^2+1)y = \tan x$: the PI stays bounded where $\tan x$ blows up. ESA ★★

ADDED FOR CLARITY · S009·5 · S009·7

How to read it. Slide 009·7 with $a = 1$. Vermilion dashed: $X = \tan x$, with asymptotes at $x = \pm\frac{\pi}{2}$ (grey). Blue: the PI $-\cos x \ln(\sec x + \tan x)$; it is odd, like $\tan x$, and tends to 0 at $x = \pm\frac{\pi}{2}$ (where $\cos x \to 0$ beats the logarithm) although X is unbounded there.

Flowcharts



Remark. The constants $-\frac{1}{a}$ cancel and $e^{iax}+e^{-iax}=2\cos ax$ makes the result real.

Figure F2.1. Slides 009·5–7: $(D^2+a^2)y=\tan ax$ ESA ★★

S009·5 · S009·6 · S009·7 · T009 00:04:12 · T009 00:11:09

tan, sec, csc right-hand sides: operator method or variation of parameters

	Operator method (this pack)	Variation of parameters (class 011)
Idea	$\frac{1}{D-m}X = e^{mx}\int Xe^{-mx}dx$, complex $m = \pm ia$	$y_p = -y_1\int\frac{y_2X}{W}dx + y_2\int\frac{y_1X}{W}dx$
Arithmetic	complex numbers, Euler's formula	real; needs the Wronskian W
Same answer?	yes: $\frac{\tan ax}{D^2+a^2} = -\frac{1}{a^2}\cos ax\log(\sec ax + \tan ax)$	yes (up to CF terms)
Asked as	slides 009·5–7; ESA Dec-23 3c (e^{e^x})	CW P23 ($\sec ax$), P26 ($\csc x$), P28 ($\tan 2x$); ESA VoP questions
When to use	question says "solve" and X is no standard form	question says "by the method of variation of parameters"

ADDED FOR CLARITY S009·5 · S009·7 · CW U2 P23 · CW U2 P26 · CW U2 P28 · ESA Jan-24 s1 3a

Important points and why

1. $D^2+a^2 = (D+ia)(D-ia)$, so $\frac{1}{D^2+a^2} = \frac{1}{2ia}\left[\frac{1}{D-ia} - \frac{1}{D+ia}\right]$. ESA ★★ S009·5 · S009·6 · T009 00:04:31
2. Compute $\frac{1}{D-ia}\tan ax$ once; get $\frac{1}{D+ia}\tan ax$ by changing i to $-i$ ("we simply replace i by $-i$"). ESA ★★ S009·7 · T009 00:08:58
3. The imaginary parts must cancel: the PI of a real equation is real ($e^{iax}+e^{-iax} = 2\cos ax$ at the end). ESA ★★ S009·7 · T009 00:10:30 · T009 00:10:48
4. Result to remember: $\frac{1}{D^2+a^2}\tan ax = -\frac{1}{a^2}\cos ax\log(\sec ax + \tan ax)$. ESA ★★ S009·7 · T009 00:11:09
5. $\tan ax$, $\sec ax$, $\csc ax$ right-hand sides can equally be solved by variation of parameters (class 011); the CW sheet asks them that way (P23, P26, P28). Read the question: "by the method of variation of parameters" forces that method. ESA ★★ ADDED FOR CLARITY CW U2 P23 · CW U2 P26 · CW U2 P28

Why is the PI real although the computation is complex?
The equation has real coefficients and a real right-hand side. The two partial results are complex conjugates, so $\frac{1}{2ia}(P_1 - \bar{P}_1) = \frac{1}{a}\mathrm{Im}\,P_1$ is real.
ADDED FOR CLARITY S009·7 · T009 00:10:30

Exam trap. Do not integrate $\frac{\sin^2 ax}{\cos ax}$ in one go: write it as $\sec ax - \cos ax$ first, then $\int\sec ax\,dx = \frac{1}{a}\log(\sec ax + \tan ax)$. ADDED FOR CLARITY S009·6 · T009 00:06:46

Exam trap. The PI has $\frac{1}{a^2}$, not $\frac{1}{a}$: one $\frac{1}{a}$ from the integration and one from $\frac{1}{2ia}$.
ADDED FOR CLARITY S009·7

Worked examples

W2.1 · $(D^2+a^2)y=\tan ax$ (slides 009·5–7) ESA ★★

GIVEN Solve $(D^2+a^2)y=\tan ax$.

STEP 1 AE $m^2+a^2=0$: $m=\pm ia$ ($\alpha=0$, $\beta=a$). CF $=c_1\cos ax+c_2\sin ax$.

STEP 2 PI $=\frac{1}{(D+ia)(D-ia)}\tan ax=\frac{1}{2ia}\left[\frac{1}{D-ia}-\frac{1}{D+ia}\right]\tan ax$.

STEP 3 $\frac{1}{D-ia}\tan ax=e^{iax}\int\tan ax\,(\cos ax-i\sin ax)\,dx=e^{iax}\int\left(\sin ax-i\frac{\sin^2 ax}{\cos ax}\right)dx$.

STEP 4 $\frac{\sin^2 ax}{\cos ax}=\sec ax-\cos ax$: the integral is $-\frac{\cos ax}{a}-\frac{i}{a}\log(\sec ax+\tan ax)+\frac{i}{a}\sin ax$ $=-\frac{1}{a}\left[(\cos ax-i\sin ax)+i\log(\sec ax+\tan ax)\right]$.

STEP 5 Times e^{iax} ($e^{iax}(\cos ax-i\sin ax)=1$): $\frac{1}{D-ia}\tan ax=-\frac{1}{a}\left[1+ie^{iax}\log(\sec ax+\tan ax)\right]$.

STEP 6 $i\to -i$: $\frac{1}{D+ia}\tan ax=-\frac{1}{a}\left[1-ie^{-iax}\log(\sec ax+\tan ax)\right]$.

STEP 7 PI $=\frac{1}{2ia}\cdot\left(-\frac{i}{a}\right)(e^{iax}+e^{-iax})\log(\sec ax+\tan ax)=-\frac{1}{a^2}\cos ax\log(\sec ax+\tan ax)$.

ANSWER $$\boxed{y=c_1\cos ax+c_2\sin ax-\frac{1}{a^2}\cos ax\,\log(\sec ax+\tan ax)}$$

CHECK Slide answer agrees; sympy: (D^2+a^2) of the PI minus $\tan ax$ simplifies to 0 (symbolic a), and both partial results satisfy $u'\mp iau=\tan ax$ numerically at $a=2$, $x=0.1, 0.3$. Figure G2.1. By variation of parameters: class 011 pack.

S009·5 · S009·6 · S009·7 · T009 00:03:13 · T009 00:11:09

SHORT-NOTE RECAP

$(D^2+a^2)y=\tan ax$: $\frac{1}{D^2+a^2}=\frac{1}{2ia}\left[\frac{1}{D-ia}-\frac{1}{D+ia}\right]$; $\frac{1}{D-ia}\tan ax=e^{iax}\int\tan ax\,e^{-iax}dx=-\frac{1}{a}[1+ie^{iax}\log(\sec ax+\tan ax)]$; $i\to -i$ for the other. PI $=-\frac{1}{a^2}\cos ax\log(\sec ax+\tan ax)$. tan, sec, csc can also be done by variation of parameters (class 011).

3 Right-hand sides in e^x: the substitution $t = e^x$

When X is a function of e^x (e^{e^x}, $\sin(e^x)$, $e^{2x}\sin(e^x)$), each integral $e^{mx}\int Xe^{-mx}\,dx$ becomes, with $t = e^x$, $dt = e^x\,dx$, an integral of a power of t times e^t or $\sin t$, done by parts (or Bernoulli's rule). The one ESA question of this kind, Dec-23 3c, is $y'' + 3y' + 2y = e^{e^x}$, the same equation as slide 009·10: its PI is $e^{-2x}e^{e^x}$. Applying the factors one after another, in the right order, is a shorter route (added for clarity).

Key terms and definitions

Substitution $t = e^x$. $dt = e^x\,dx$, so $e^{kx}\,dx = t^{k-1}\,dt$; it turns $\int e^{kx}F(e^x)\,dx$ into $\int t^{k-1}F(t)\,dt$. S009·11 · S009·12 · S009·15 · T009b 00:03:57

Bernoulli's rule. Repeated integration by parts in one line: $\int uv\,dx = uv_1 - u'v_2 + u''v_3 - \cdots$, where $v_1, v_2, \ldots$ are successive integrals of v; used for $\int t^3\sin t\,dt$, $\int t^4\sin t\,dt$. T009b 00:08:37 · S009·16

Formulas

E3.1 · The substitution $t = e^x$ ESA ★★

$$e^{-mx}\int e^{kx}F(e^x)\,dx = e^{-mx}\int t^{k-1}F(t)\,dt, \qquad t = e^x,\ dt = e^x dx$$

gives	Integrals of powers of e^x times a function of e^x.
applies	$X = e^{e^x}$, $\sin(e^x)$, $e^{2x}\sin(e^x)$, ...
note	Write $e^{kx}dx = e^{(k-1)x}\cdot e^x dx = t^{k-1}dt$; substitute back $t = e^x$ at the end.

S009·11 · S009·12 · S009·13 · S009·15 · T009 00:14:05 · T009b 00:03:57 · ESA Dec-23 3c

E3.2 · Integrals in t ESA ★★

$$\int te^t\,dt = (t-1)e^t, \qquad \int t\sin t\,dt = -t\cos t + \sin t$$

$$\int t^3\sin t\,dt = -t^3\cos t + 3t^2\sin t + 6t\cos t - 6\sin t$$

gives	The by-parts results used on slides 009·11, 009·13, 009·16.
applies	After $t = e^x$.
note	$\int t^4\sin t\,dt = -t^4\cos t + 4t^3\sin t + 12t^2\cos t - 24t\sin t - 24\cos t$.

S009·11 · S009·13 · S009·16 · T009 00:14:43 · T009b 00:05:16 · T009b 00:08:37

E3.3 · ESA Dec-23 3c: PI of e^{e^x} ESA ★★

$$\frac{1}{D^2+3D+2}\,e^{e^x} = e^{-2x}e^{e^x}$$

GIVES The PI; $y = c_1e^{-x} + c_2e^{-2x} + e^{-2x}e^{e^x}$.

APPLIES $y'' + 3y' + 2y = e^{e^x}$.

NOTE Same as slide 009·10–11; checked by substitution (sympy).

S009·10 · S009·11 · ESA Dec-23 3c

Graphs and diagrams

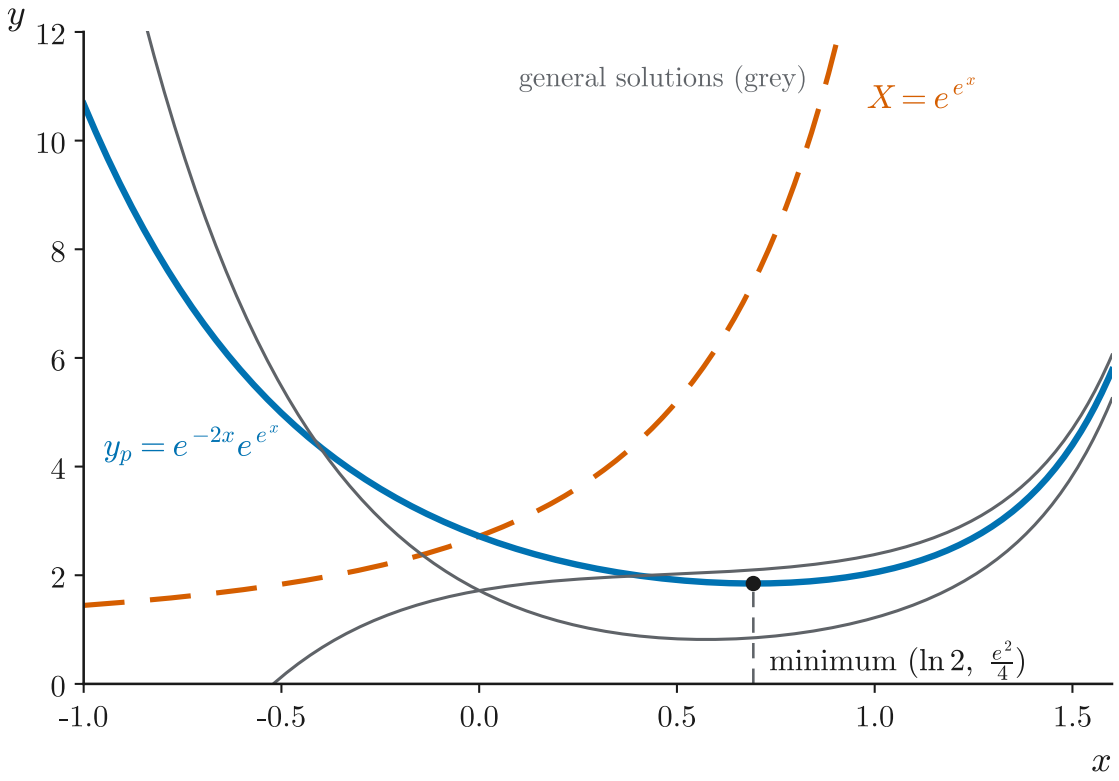


Figure G3.1. ESA Dec-23 3c: $y_p = e^{-2x}e^{e^x}$ and general solutions. ESA ★★

ADDED FOR CLARITY ESA Dec-23 3c · S009·11

How to read it. Vermilion dashed: $X = e^{e^x}$. Blue: the PI $e^{-2x}e^{e^x}$; its minimum is where $(e^x - 2x)' = e^x - 2 = 0$, at $x = \ln 2$, value $\frac{e^2}{4}$. Grey: two general solutions $c_1e^{-x} + c_2e^{-2x} + y_p$ ($c_1, c_2 = 2, -3$ and $-3, 2$); the CF dies out for large x, so every solution approaches y_p there; note $y_p = e^{-2x}X$ exactly.

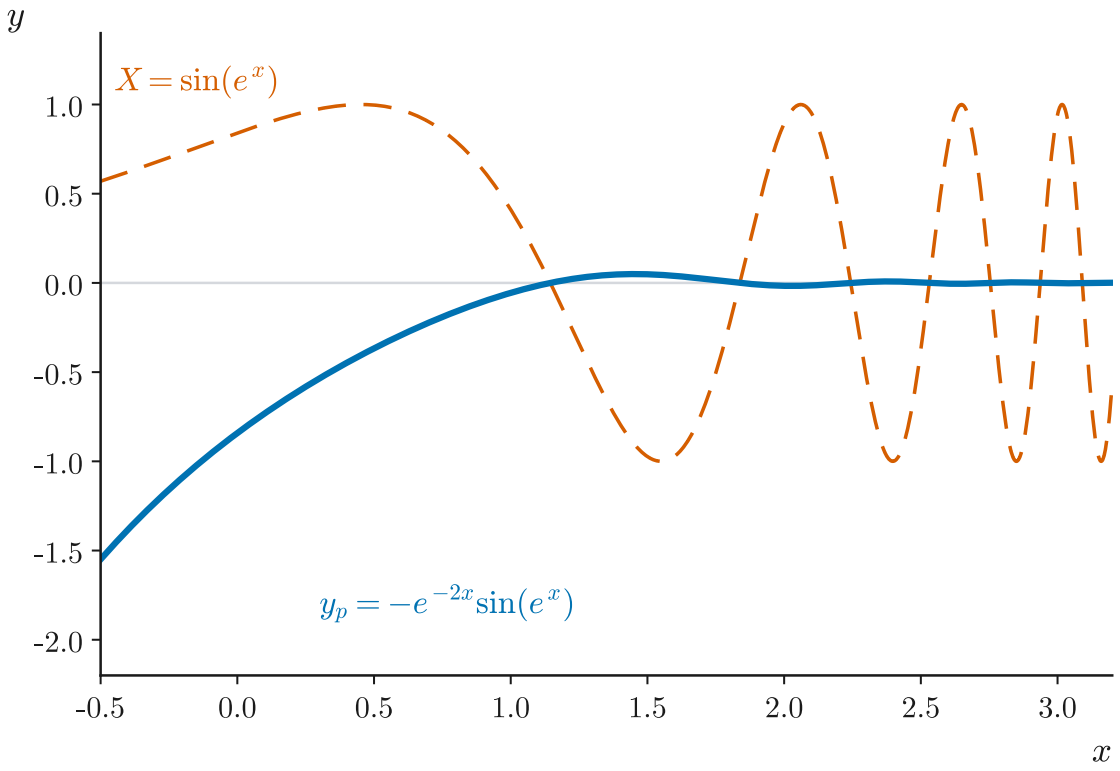
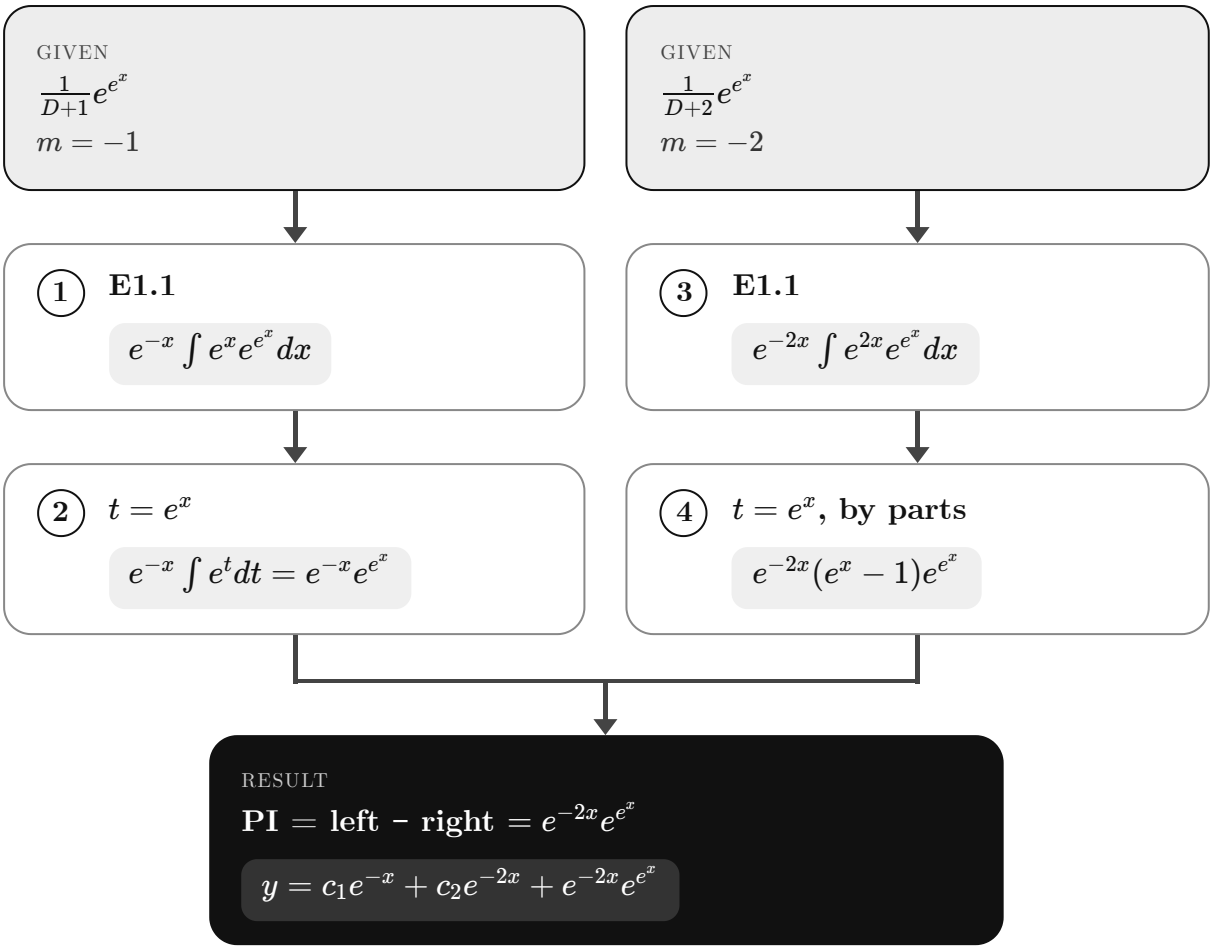


Figure G3.2. $(D+1)(D+2)y = \sin(e^x)$: a chirp in, a damped chirp out. ESA ★★

ADDED FOR CLARITY S009·12 · S009·13

How to read it. Vermilion dashed: $X = \sin(e^x)$, which oscillates faster and faster as x grows. Blue: the PI $-e^{-2x}\sin(e^x)$: the same zeros ($e^x = k\pi$) but an amplitude e^{-2x} that dies away, because $f(D)$ amplifies fast oscillations (D^2 dominates), so their PI is small.

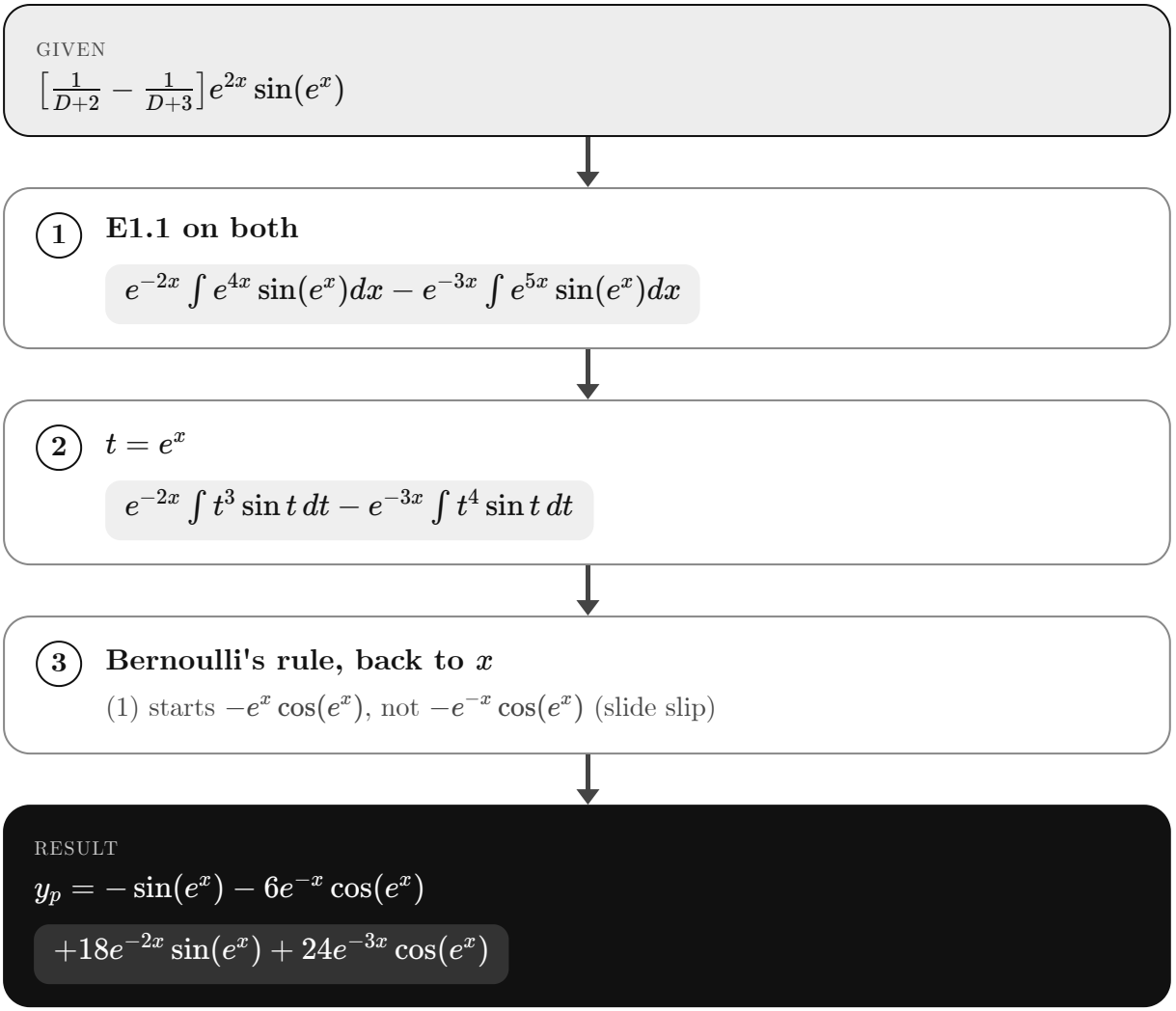
Flowcharts



Remark. CF from $m=-1,-2$. Worked in full as W3.1 (7 marks).

Figure F3.1. ESA Dec-23 3c: $y''+3y'+2y=e^{e^x}$ ESA ★★

ESA Dec-23 3c · S009·10 · S009·11



Exam trap. Slide 009·16's line (1) has $-e^{-x}\cos(e^x)$; with that slip the e^x terms would not cancel.

Figure F3.2. Slides 009·14–17: $(D+2)(D+3)y = e^{2x}\sin(e^x)$ ESA ★★

S009·14 · S009·15 · S009·16 · S009·17

Every "any other form" problem in this pack

Source (worked)	Equation	Integrals after $t = e^x$ / by parts	PI
S009·8–9 (W1.1)	$D^2y = \ln x$	$\iint \ln x$	$\frac{x^2 \ln x}{2} - \frac{3x^2}{4}$
S009·5–7 (W2.1)	$(D^2 + a^2)y = \tan ax$	$\int \sin ax$, $\int \sec ax$	$-\frac{1}{a^2}\cos ax \log(\sec ax + \tan ax)$
S009·10–11, ESA (W3.1)	$(D^2 + 3D + 2)y = e^{e^x}$	$\int e^t$, $\int te^t$	$e^{-2x}e^{e^x}$
S009·12–13 (W3.2)	$(D + 1)(D + 2)y = \sin(e^x)$	$\int \sin t$, $\int t \sin t$	$-e^{-2x}\sin(e^x)$
S009·14–17 (W3.3)	$(D^2 + 5D + 6)y = e^{2x}\sin(e^x)$	$\int t^3 \sin t$, $\int t^4 \sin t$	$-\sin(e^x) - 6e^{-x}\cos(e^x) + \ldots$

ADDED FOR CLARITY S009·5 · S009·8 · S009·10 · S009·12 · S009·14 · ESA Dec-23 3c

Important points and why

1. For X built from e^x, substitute $t = e^x$, $dt = e^x dx$ in each integral: $e^{kx}dx = t^{k-1}dt$. ESA ★★ S009·11 · S009·12 · S009·15 · T009 00:14:05 · T009b 00:03:57
2. Split the exponential to make $e^x dx$ visible: $e^{2x} = e^x \cdot e^x$, $e^{4x} = e^{3x} \cdot e^x$ ("let us split e^{2x} to be e^x times e^x"). ESA ★★ T009 00:14:22 · T009b 00:04:57 · T009b 00:08:00
3. ESA Dec-23 3c ($y'' + 3y' + 2y = e^{e^x}$, 7 marks) is the slide 009·10 problem: PI $= e^{-2x}e^{e^x}$; write the full y. ESA ★★ ESA Dec-23 3c · S009·10 · S009·11
4. Always simplify after substituting back: in $\frac{1}{(D+1)(D+2)}\sin(e^x)$ the terms $-e^{-x}\cos(e^x)$ cancel, leaving $-e^{-2x}\sin(e^x)$. ESA ★★ S009·13 · T009b 00:06:04
5. Choosing the order $\frac{1}{D+2}\left[\frac{1}{D+1}X\right]$ for e^{e^x} and $\sin(e^x)$ needs only $\int e^t dt$ and $\int \cos t\, dt$: two lines each (W3.1, W3.2). ESA ★★ ADDED FOR CLARITY S009·10 · S009·12 · ESA Dec-23 3c

Why does the order of successive factors matter for e^{e^x}?
$\frac{1}{D+1}e^{e^x} = e^{-x}e^{e^x}$ is easy, and then $\frac{1}{D+2}$ needs $\int e^{2x}e^{-x}e^{e^x}dx = \int e^x e^{e^x}dx = e^{e^x}$. The other order leaves $\int(1 - e^{-x})e^{e^x}dx$, which needs spotting a derivative.
ADDED FOR CLARITY S009·10 · ESA Dec-23 3c

Exam trap. Slide 009·16 (and 009·17) writes the first term of (1) as $-e^{-x}\cos(e^x)$; it is $e^{-2x} \cdot (-e^{3x}\cos(e^x)) = -e^x\cos(e^x)$. The boxed answer is right only because the correct $-e^x\cos(e^x)$ cancels with (2). ADDED FOR CLARITY S009·16 · S009·17

Exam trap. After $t = e^x$, also convert the dx: $e^{2x}e^{e^x}dx = te^t\,dt$, not $t^2e^t\,dt$ (one e^x goes into dt).
ADDED FOR CLARITY S009·11 · T009 00:14:22

Worked examples

W3.1 · $y'' + 3y' + 2y = e^{e^x}$ (ESA Dec-23 3c; slides 009·10–11) ESA ★★

GIVEN Solve $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = e^{e^x}$, i.e. $(D^2 + 3D + 2)y = e^{e^x}$ (7 marks).

STEP 1 AE $m^2 + 3m + 2 = 0$: $(m+1)(m+2) = 0$, $m = -1, -2$. CF $= c_1e^{-x} + c_2e^{-2x}$.

STEP 2 e^{e^x} is none of the standard forms. PI $= \frac{1}{(D+1)(D+2)}e^{e^x} = \left[\frac{1}{D+1} - \frac{1}{D+2}\right]e^{e^x}$.

STEP 3 $\frac{1}{D+1}e^{e^x} = e^{-x}\int e^x e^{e^x}dx$; $t = e^x$, $dt = e^x dx$: $e^{-x}\int e^t dt = e^{-x}e^{e^x}$.

STEP 4 $\frac{1}{D+2}e^{e^x} = e^{-2x}\int e^{2x}e^{e^x}dx = e^{-2x}\int te^t dt = e^{-2x}(t-1)e^t = e^{-2x}(e^x - 1)e^{e^x}$.

STEP 5 PI $= e^{-x}e^{e^x} - e^{-x}e^{e^x} + e^{-2x}e^{e^x} = e^{-2x}e^{e^x}$.

STEP 6 Shortcut (E1.4): $\frac{1}{D+2}\left[\frac{1}{D+1}e^{e^x}\right] = \frac{1}{D+2}\left[e^{-x}e^{e^x}\right] = e^{-2x}\int e^x e^{e^x}dx = e^{-2x}e^{e^x}$.

ANSWER $\boxed{y = c_1e^{-x} + c_2e^{-2x} + e^{-2x}e^{e^x}}$

CHECK Slide 009·11's PI agrees; sympy dsolve gives $y = (c_1 + (c_2 + e^{e^x})e^{-x})e^{-x}$, the same, and checkodesol returns 0. Figure G3.1, Figure F3.1.

ESA Dec-23 3c · S009·10 · S009·11 · T009 00:13:23 · T009 00:14:43

W3.2 · $(D+1)(D+2)y = \sin(e^x)$ (slides 009·12–13) ESA ★★

GIVEN Find the particular integral of $(D+1)(D+2)y = \sin(e^x)$.

STEP 1 (CF, for the general solution: $c_1e^{-x} + c_2e^{-2x}$.)

STEP 2 $y_p = \left[\frac{1}{D+1} - \frac{1}{D+2}\right]\sin(e^x)$.

STEP 3 $\frac{1}{D+1}\sin(e^x) = e^{-x}\int e^x\sin(e^x)dx = e^{-x}\int \sin t\,dt = -e^{-x}\cos(e^x)$.

STEP 4 $\frac{1}{D+2}\sin(e^x) = e^{-2x}\int t\sin t\,dt = e^{-2x}(-t\cos t + \sin t) = -e^{-x}\cos(e^x) + e^{-2x}\sin(e^x)$.

STEP 5 $y_p = -e^{-x}\cos(e^x) - \left[-e^{-x}\cos(e^x) + e^{-2x}\sin(e^x)\right] = -e^{-2x}\sin(e^x)$.

STEP 6 Shortcut: $\frac{1}{D+2}\left[-e^{-x}\cos(e^x)\right] = -e^{-2x}\int e^x\cos(e^x)dx = -e^{-2x}\sin(e^x)$.

ANSWER $\boxed{y_p = -e^{-2x}\sin(e^x);\ y = c_1e^{-x} + c_2e^{-2x} - e^{-2x}\sin(e^x)}$

CHECK Slide answer agrees; sympy dsolve gives the same; substituting back leaves 0. Figure G3.2.

S009·12 · S009·13 · T009b 00:01:25 · T009b 00:06:24

W3.3 · $(D^2+5D+6)y = e^{2x}\sin(e^x)$ (slides 009·14–17) ESA ★★

GIVEN Find the particular integral of $(D^2+5D+6)y = e^{2x}\sin(e^x)$.

STEP 1 $(D+2)(D+3)$; CF $= c_1e^{-2x} + c_2e^{-3x}$. $y_p = \left[\frac{1}{D+2} - \frac{1}{D+3}\right]e^{2x}\sin(e^x)$.

STEP 2 $= e^{-2x}\int e^{4x}\sin(e^x)dx - e^{-3x}\int e^{5x}\sin(e^x)dx = e^{-2x}\int t^3\sin t\,dt - e^{-3x}\int t^4\sin t\,dt$.

STEP 3 (1) $e^{-2x}[-t^3\cos t + 3t^2\sin t + 6t\cos t - 6\sin t] = -e^x\cos(e^x) + 3\sin(e^x) + 6e^{-x}\cos(e^x) - 6e^{-2x}\sin(e^x)$. **Slide 009·16 prints the first term as $-e^{-x}\cos(e^x)$.**

STEP 4 (2) $e^{-3x}[-t^4\cos t + 4t^3\sin t + 12t^2\cos t - 24t\sin t - 24\cos t] = -e^x\cos(e^x) + 4\sin(e^x) + 12e^{-x}\cos(e^x) - 24e^{-2x}\sin(e^x) - 24e^{-3x}\cos(e^x)$.

STEP 5 (1) − (2): the $-e^x\cos(e^x)$ terms cancel; $3-4=-1$, $6-12=-6$, $-6+24=18$, $0+24=24$.

ANSWER $\boxed{y_p = -\sin(e^x) - 6e^{-x}\cos(e^x) + 18e^{-2x}\sin(e^x) + 24e^{-3x}\cos(e^x)}$

CHECK Slide's boxed answer agrees (despite the slip in line (1)); sympy: $\int t^3\sin t$, $\int t^4\sin t$ as above, (1)−(2) equals y_p, dsolve gives the same PI, and substituting back leaves 0.

S009·14 · S009·15 · S009·16 · S009·17 · T009b 00:06:50 · T009b 00:09:35

Past-paper questions

ESA Dec 2023 (7 marks) UE22MA141A Q3.c · solved in W3.1

Solve $y'' + 3y' + 2y = e^{e^x}$.

SHORT-NOTE RECAP

X in e^x: $t = e^x$, $e^{kx}dx = t^{k-1}dt$. ESA Dec-23 3c: $\frac{e^{e^x}}{(D+1)(D+2)} = e^{-x}e^{e^x} - e^{-2x}(e^x-1)e^{e^x} = e^{-2x}e^{e^x}$, $y = c_1e^{-x} + c_2e^{-2x} + e^{-2x}e^{e^x}$. $\frac{\sin(e^x)}{(D+1)(D+2)} = -e^{-2x}\sin(e^x)$. $\frac{e^{2x}\sin(e^x)}{(D+2)(D+3)} = -\sin(e^x) - 6e^{-x}\cos(e^x) + 18e^{-2x}\sin(e^x) + 24e^{-3x}\cos(e^x)$.

A Symbol glossary

Symbol	Name	Meaning	Appears in
D	Differential operator	$\frac{d}{dx}$; $\frac{1}{D}$ = integration without a constant	E1.1, E1.5
$f(D)$	Operator polynomial	Left-hand operator of $f(D)y = X$	E1.3
X	Right-hand side	A function of x (capital); here not of a standard form	E1.1
m_k	AE root	Root of $f(m) = 0$; $f(D) = \prod(D - m_k)$	E1.1, E1.3
A_k	Partial-fraction coefficient	$\frac{1}{f(D)} = \sum \frac{A_k}{D - m_k}$; $A_k = \frac{1}{f'(m_k)}$ (not an arbitrary constant)	E1.3
a	Constant	In $D^2 + a^2$, $\tan ax$: $m = \pm ia$; in the lecture's $\frac{1}{D-a} f(x)$ it is the root (clash)	E2.1, E2.5
i	Imaginary unit	$i^2 = -1$; $e^{\pm iax} = \cos ax \pm i \sin ax$	E2.2, E2.4
t	Substitution variable	$t = e^x$, $dt = e^x dx$ (not time)	E3.1, E3.2
$\ln x$, $\log x$	Natural logarithm	Both mean $\log_e x$ on these slides	E1.6, E2.3
c_1, c_2	Arbitrary constants	In the CF only	E2.5, E3.3

B Sources, coverage and gaps

Source	Path	Use
Deck 009 (.pptx rendered; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/009_Particular_integral_when_X_is_any_other_form__0611074c2202edfa.*	18 slides; content on 3–17 (every equation read from the rendered images)
Transcript T009 (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/009_Particular_Integral_when_X_is_any_other_form__653ae8f75a03fb1e.*	15.2 min; slides 3–11 (method, $\tan ax$, $\ln x$, e^{e^x})
Transcript T009b (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/009_Particular_integral_when_X_is_any_other_form_Continued__5c24666421028690.*	10.2 min; slides 12–17 ($\sin(e^x)$, $e^{2x}\sin(e^x)$)
Unit 1 & 2 class-work sheet (UE24MA141A)	pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf	Only as a pointer: P23 ($\sec ax$), P26 ($\csc x$), P28 ($\tan 2x$) are variation-of-parameters problems (class 011)
ESA papers, UE22MA141A and UE23MA141A (8 papers)	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE2[23]MA141A/*.pdf	Dec-23 3c, via the shared ESA map
Video 009 (.mp4, both parts)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/009_*.mp4	Not opened: every slide equation is readable and no video trigger applies

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024 (19 distinct Unit 2 questions, 24 occurrences). This session's method is asked exactly once: ESA Dec-23 3c, $y'' + 3y' + 2y = e^{e^x}$ (7 marks), never repeated in the later papers; it is the same equation as slide 009·10 (W3.1). Stars: partial fractions with $\frac{1}{D-m}X$ and the substitution $t = e^x$ ★★ (one ESA question); $\tan ax$ ★★ (its answer with $a = 2$ is CW P28, which the sheet asks by variation of parameters); $\frac{1}{D^2}\ln x$ and the complex-factor details ★. No CW problem is assigned to this session: the CW sheet's tan, sec, csc problems (P23, P26, P28) are in the variation-of-parameters section and are solved in the class 011 VoP pack.

Known gaps, conflicts and corrections

1. Slide 009·16, line (1), writes the first term of $e^{-2x}\int t^3 \sin t\, dt$ as $-e^{-x}\cos(e^x)$; it is $e^{-2x}(-t^3\cos t) = -e^x\cos(e^x)$. Slide 009·17 copies the slip into its first bracket. The boxed answer $y_p = -\sin(e^x) - 6e^{-x}\cos(e^x) + 18e^{-2x}\sin(e^x) + 24e^{-3x}\cos(e^x)$ is nevertheless right: with the correct $-e^x\cos(e^x)$ the two leading terms of (1) and (2) cancel (sympy; W3.3).

2. Every other slide line was recomputed with sympy: 009·6–7 ($\frac{1}{D\mp ia}\tan ax$ and the PI $-\frac{1}{a^2}\cos ax\log(\sec ax+\tan ax)$), 009·9 ($\frac{x^2\ln x}{2}-\frac{3x^2}{4}$), 009·10–11 ($A=-1$, $B=1$, PI $e^{-2x}e^{e^x}$), 009·12–13 ($-e^{-2x}\sin(e^x)$) and 009·16's $\int t^3\sin t\,dt$, $\int t^4\sin t\,dt$ are right; the AE roots on 009·5 ($\pm ia$), 009·8 $(0,0)$ agree with their CFs. Slide 009·8 prints "$m^2=0\Rightarrow\Rightarrow m=0,0$" (a doubled arrow, harmless).
3. Slides 009·10–17 ask only for "the particular integral"; the general solutions are added here (CF from the factors). ESA Dec-23 3c asks to "solve", so W3.1 gives $y=c_1e^{-x}+c_2e^{-2x}+e^{-2x}e^{e^x}$.
4. Applying the linear factors one after another, without partial fractions (E1.4), is not on the slides; it is added for clarity because for e^{e^x} and $\sin(e^x)$ the order $\frac{1}{D+2}\left[\frac{1}{D+1}X\right]$ needs only one easy integral each.
5. T009 00:15:01 closes with "these are the four standard problems"; T009 works three ($\tan ax$, $\ln x$, e^{e^x}) and T009b two more. T009 00:12:09 says "unfortunately we can solve this by a normal method"; the sense is "fortunately": $\frac{1}{D^2}\ln x$ is just two integrations. T009 00:06:46 writes $\sin^2 ax=1-\cos^2 ax$, so $\frac{\sin^2 ax}{\cos ax}=\sec ax-\cos ax$; the slide keeps $\frac{\sin^2 ax}{\cos ax}$ in one piece. Both give the same integral.
6. The CW sheet has no problem for this session. Its $\sec ax$, $\csc x$ and $\tan 2x$ problems (P23, P26, P28) say "solve by method of variation of parameters only" and belong to the class 011 VoP pack; P28's printed answer, $-\frac{1}{4}\cos 2x\log(\sec 2x+\tan 2x)$ plus CF, is slide 009·7's result with $a=2$.
7. Transcript recognition errors were corrected silently: "T square plus a square", "d square plus a square" → D^2+a^2; "axillary", "axlary" → auxiliary; "1 over e minus a", "1 over p minus 1 of f of x" → $\frac{1}{D-a}f(x)$; "second ax" → $\sec ax$; "than ax" → $\tan ax$; "car square ax" → $\cos^2 ax$; "interlog x" → into $\log x$; "t2x" → c_2x; "EE power x", "sine of v power x", "e polix" → e^{e^x}, $\sin(e^x)$, e^x; "psi t", "t s t dt" → $\sin t$, $t\sin t\,dt$; "phi d plus six" → $5D+6$; "we become plus one" → B becomes $+1$; "area is minus minus 1" → a is -1; "absorb it" → observe; "by partnoli rule" → by parts or Bernoulli's rule; "Sol the" → Solve the.

Coverage

Source	Citations	Slides cited (slide × citations)
S009	113	3×8, 4×9, 5×8, 6×8, 7×11, 8×6, 9×6, 10×11, 11×13, 12×11, 13×5, 14×4, 15×5, 16×5, 17×3
CW	7	
ESA	15	
T009	51	
T009b	23	

The deck's own title slide says "SESSION: 7"; the bucket numbers it 009 (bucket numbers are not contiguous: the decks say SESSION 1–11; 007, 008, 010, 012 and 013 do not exist; 009 = session 7, 011 = sessions 8 (VoP) and 9 (Legendre), 014 = 10, 015 = 11). Slides 1, 2 and 18 are title and closing slides; slides 3–4 are the method; slides 5–7 $\tan ax$; 8–9 $\ln x$; 10–11 e^{e^x}; 12–13 $\sin(e^x)$; 14–17 $e^{2x}\sin(e^x)$. T009 covers slides 3–11 and T009b slides 12–17, so every content slide has a lecture. No CW problem belongs to this session (the sheet has no "any other form" section).